Indian Institute of Information Technology Design and Manufacturing, Kurnool

Department of Electronics and Communication Engineering
Bachelor of Technology

Project Report

Solar Powered Electric Fence

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1. About the Project

1.1 Project Aim

The Solar Electric Fence project aims to develop a low-cost, efficient, and renewable energy-powered fencing system. This system is designed to protect agricultural crops and livestock by delivering controlled electric pulses along a fence line. Unlike traditional fencing systems which require mains power or manual management, the solar electric fence harnesses solar energy to operate off-grid, making it ideal for rural and remote locations.

1.2 Components and Their Roles

- **Solar Panel:** Converts sunlight into electrical energy (DC), which powers the entire system and charges the battery.
- **Battery (12V 7Ah):** Stores electrical energy generated by the solar panel to supply continuous power even during night time or low sunlight.
- **LM317L Adjustable Voltage Regulator:** Ensures the battery is charged safely with astable and adjustable voltage, protecting it from Voltage fluctuations.
- **N-Channel HEXFET Power MOSFETs (IRF3205):** Act as electronic switches that drive high-current pulses from the battery into the transformer, enabling efficient pulse generation.
- **Transformer:** Steps up the low voltage pulses from MOSFETs to high voltage pulses required for energizing the fence wire.
- **CD4047 IC:** Functions as a pulse generator (astable multi vibrator) to create timing signals that control the switching of the MOSFETs, thereby dictating the pulse repetition rate.

- **Schottky Diodes:** Protect the circuit by preventing reverse current flow, especially during switching and charging phases, owing to their low forward voltage drop and fast switching capability.
- **Resistors:** Used throughout the circuit for current limiting, voltage division, setting timing parameters, and biasing components.
- **Polar and Non-Polar Capacitors:** Used for filtering power ripple, timing in multi vibrator circuits, energy storage, and reducing electrical noise.
- **LED Indicators:** Provide visual feedback on power status, charging state, and pulse generation, facilitating easy monitoring.
- **Switches:** Allow manual control to enable or disable charging and fence energization.

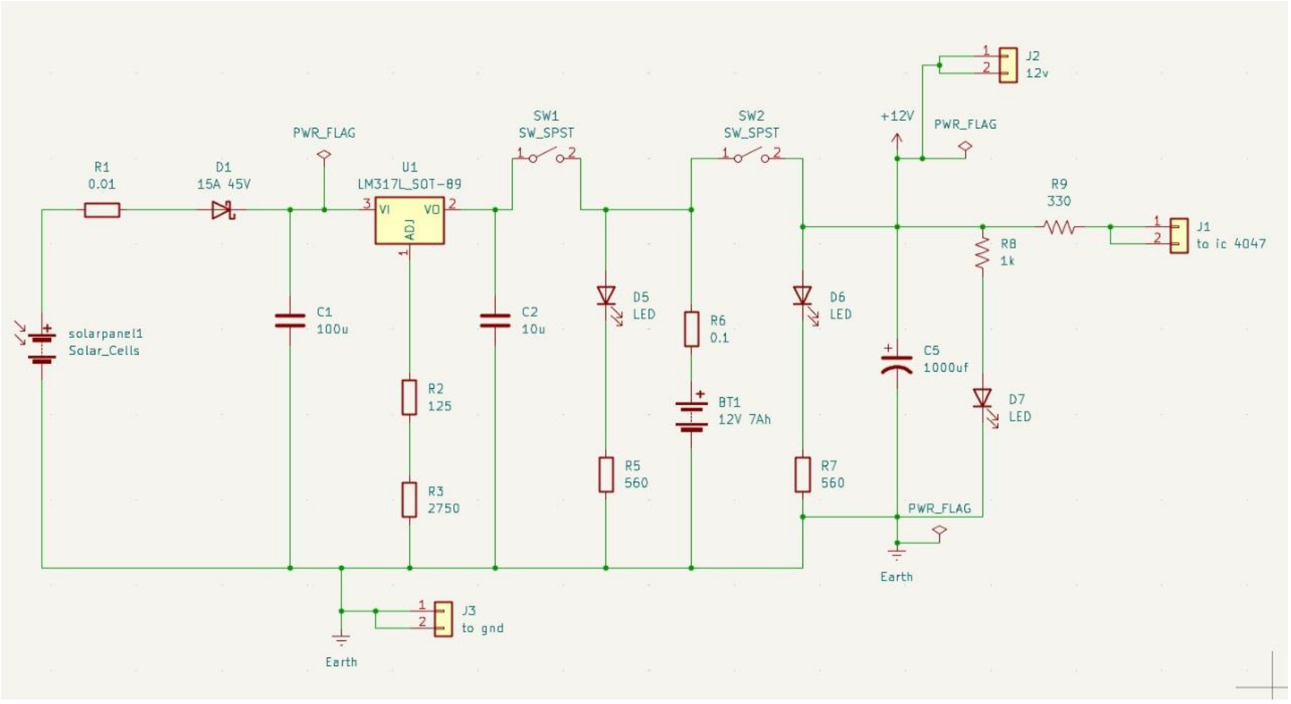
2. Project Schematics

2.1 Overview

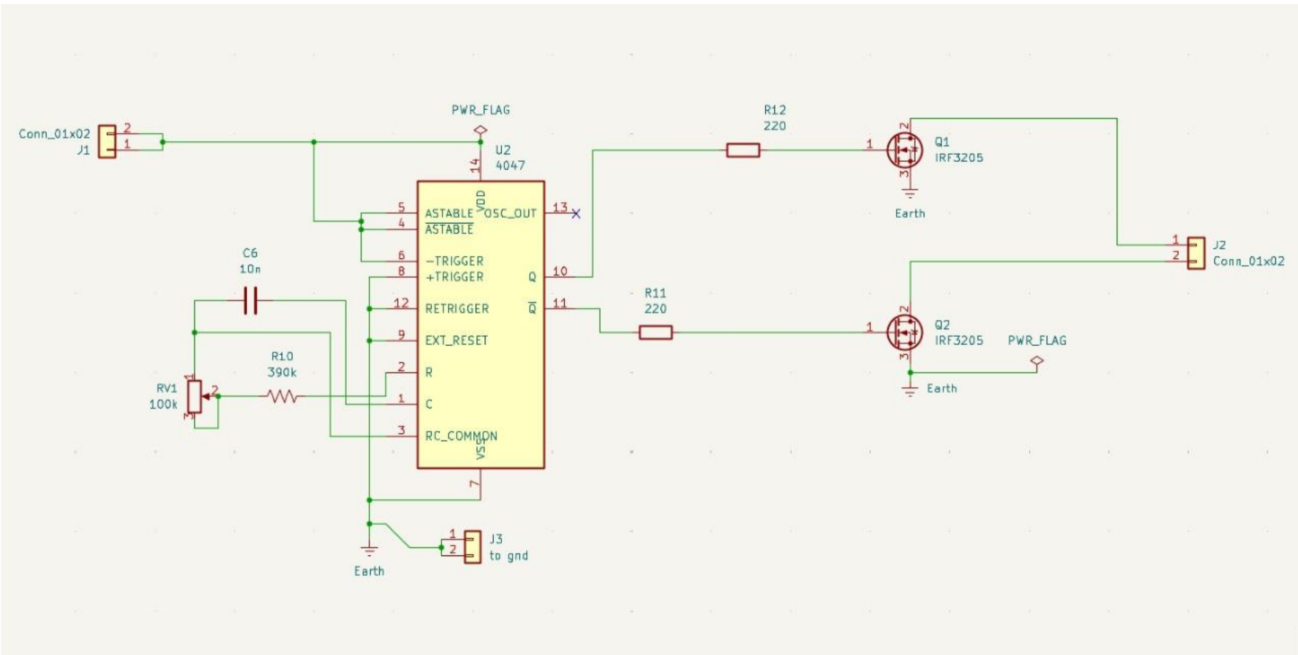
The schematic is divided into three major sections:

- **Power Supply and Charging Circuit:** Includes the solar panel input, diode protection, LM317L voltage regulator, charging control switches, and battery storage.
- **Pulse Generation Circuit:** Built around the CD4047 astable multi vibrator IC which generates a square wave to control the gate of MOSFETs.
- **Pulse Amplification and Transformer Section:** MOSFETs switch the battery power through the transformer primary, stepping voltage up to deliver sharp pulses on the fence wire.

BOARD 1


R1
0.01
D1
15A 45V
PWR_FLAG
U1
LM317L_SOT-89
VI
VO
ADJ
SW1
SW_SPST
SW2
SW_SPST
+12V
PWR_FLAG
J2
12v
R9
330
J1
to ic 4047
R8
1k
solarpanel1
Solar_Cells
C1
100u
C2
10u
D5
LED
R6
0.1
D6
LED
C5
1000uf
R2
125
BT1
12V 7Ah
D7
LED
R5
560
R7
560
R3
2750
PWR_FLAG
Earth
J3
to gnd
Earth

BOARD 2


PWR_FLAG
R12
220
Conn_01x02
J1
U2
4047
Q1
IRF3205
VDD
ASTABLE
OSC_OUT
ASTABLE
Earth
−TRIGGER
+TRIGGER
Q
J2
Conn_01x02
C6
10n
RETRIGGER
R11
220
EXT_RESET
Q2
IRF3205
PWR_FLAG
R10
390k
R
RV1
100k
C
Earth
RC_COMMON
VSS
J3
to gnd
Earth

BOARD 3

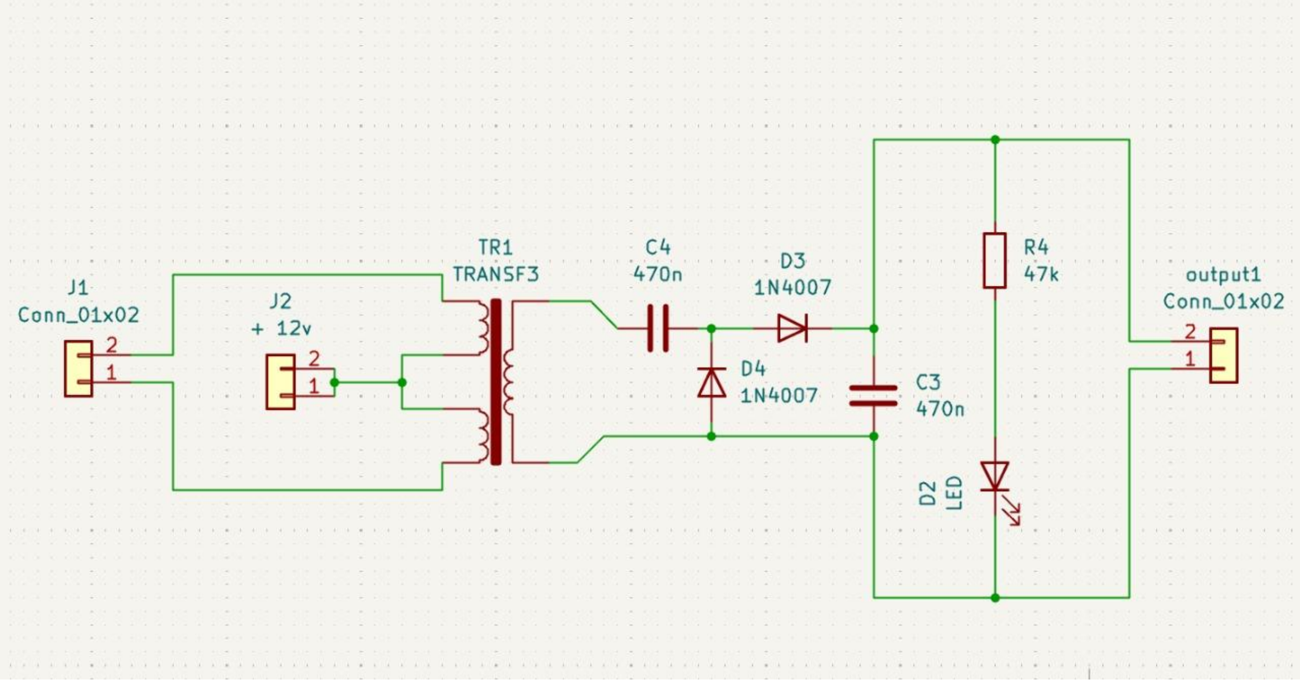


Figure : Schematic diagram of the Solar power Electric Fence

The schematic of the solar-powered electric fence circuit integrates all the essential power generation and pulse delivery stages, as summarized below:

- The solar panel interfaces with the battery charging circuit through diodes and capacitors to store energy, while switches and fuses regulate power input for safe operation.
- The 555 timer ICs provide pulse generation and triggering signals to drive the MOSFETs and transformer, enabling high-voltage output for fence energization.
- An ignition coil and transformer are used to step up the voltage from the 12V battery to deliver high-voltage pulses, ensuring effective deterrence along the fence wire.
- Supporting components such as resistors, capacitors, and LEDs are utilized for timing, filtering, status indication, and protection against electrical surges and faults.

3. Working Principle

The core functioning of the solar electric fence depends on generating high-voltage electric pulses at regulated intervals:

1. The solar panel converts sunlight into DC electricity, charging the 12V battery through an LM317L voltage regulator to avoid overcharging.
2. The CD4047 IC is configured as an astable multi vibrator producing a continuous square wave used to switch MOSFETs alternately.
3. Each MOSFET switches the battery voltage through a center - tapped transformer primary winding, causing the transformer secondary to produce high-voltage pulses.
4. These pulses are transmitted along the fence wires. When an animal contacts the fence, the circuit completes briefly, delivering a safe but discomforting shock, thereby deterring intrusion.
5. LEDs and other indicators provide the user with system status, including battery charge and output pulse activity.

Pulse Timing Calculation

The pulse frequency, f, generated by the CD4047 in astable mode can
be approximated by

$$f = \frac{1}{2.48\, R\, C}$$

where R and C are the timing resistor and capacitor connected externally to the IC. Adjusting these values changes the pulse repetition rate.

4. Design Considerations

4.1 Electrical Considerations

Careful voltage and current regulation protects battery longevity. Efficient MOSFETs minimize power loss during switching. Use of Schottky diodes reduces voltage drop and improves circuit response times.

4.2 Layout and Wiring

Short, thick wires and neat PCB layout reduce resistance and interference. Proper placement of decoupling capacitors near active devices ensures stability.

4.3 Mechanical and Environmental

Components and wiring should be weatherproofed for outdoor use. Adequate spacing and mounting support thermal dissipation and protect parts against mechanical stress.

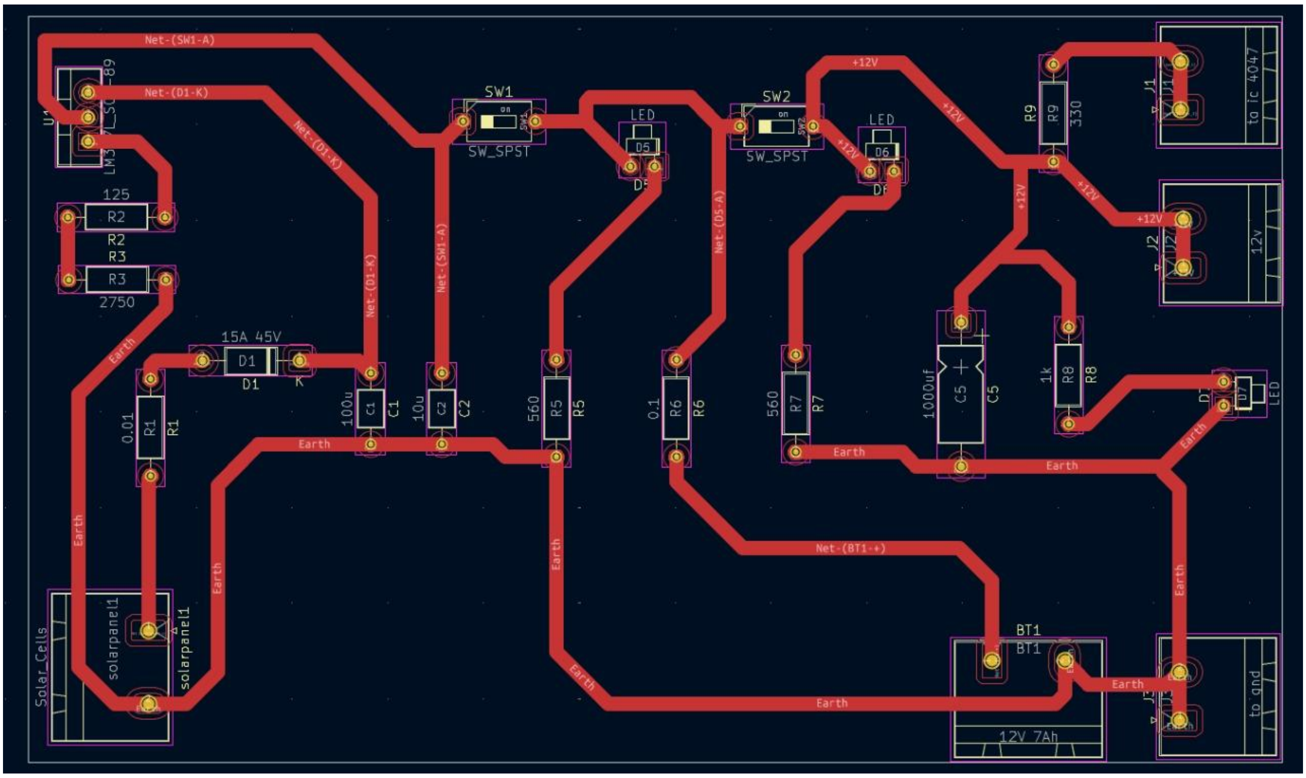
4.4 Thermal Management

High-current MOSFETs and voltage regulators should have heatsinks. Sufficient airflow and isolation prevent overheating and increase reliability.

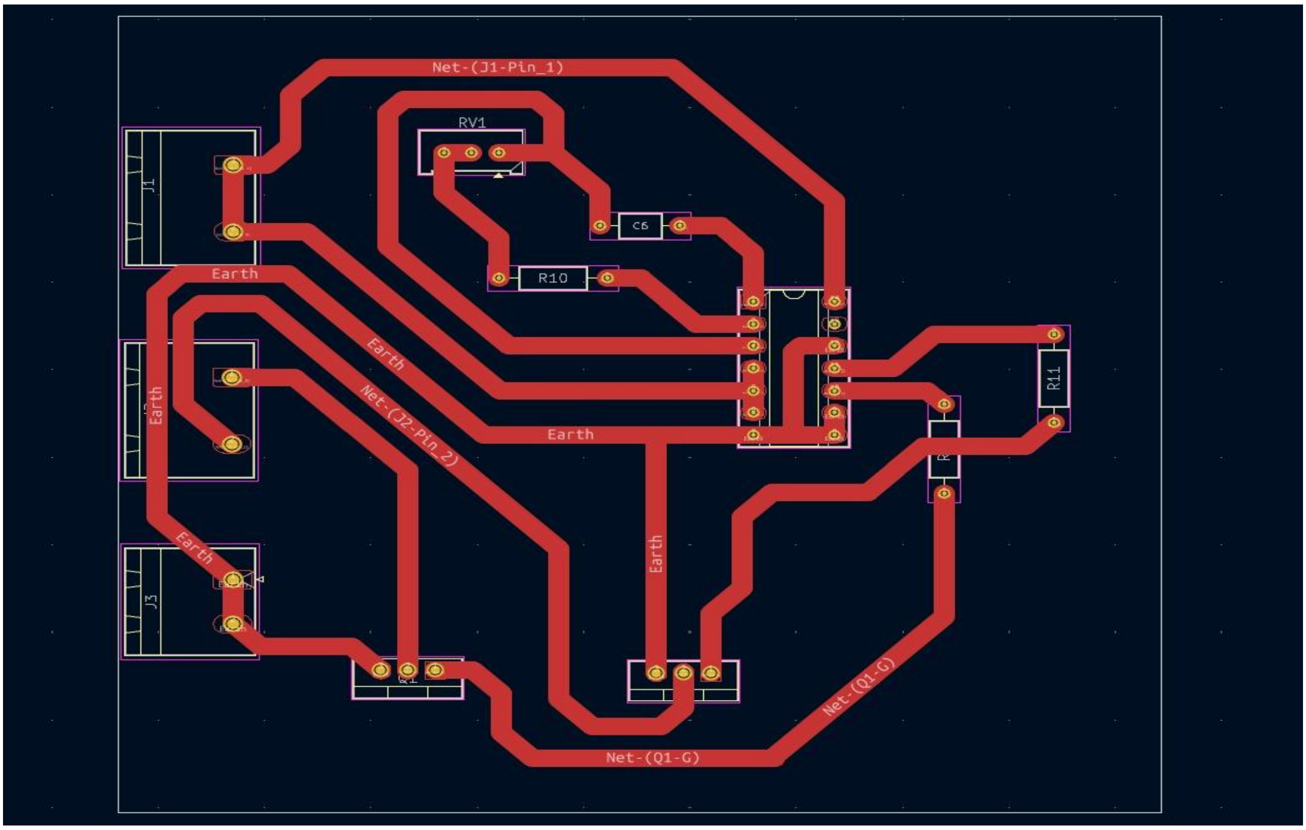
5. Layout

The PCB layout has been designed for compactness, signal integrity, and ease of assembly. The layout optimizes routing of power and signal lines, placing critical components like the voltage regulator and MOSFETs to minimize noise and thermal hotspots.

BOARD 1


Net-(SW1-A)
Net-(D1-K)
SW1
SW_SPST
LED
D5
SW2
+12V
D6
R9
330
J1
to ic 4047
J2
12v
125
R2
R3
2750
15A 45V
D1
K
0.01
R1
100u
C1
10u
C2
560
R5
0.1
R6
R7
1000uf
C5
1k
R8
LED
Earth
Net-(D5-A)
Net-(BT1-+)
BT1
12V 7Ah
to gnd
Solar_Cells
solarpanel1

BOARD 2


Net-(J1-Pin_1)
RV1
C6
R10
J1
Earth
Net-(J2-Pin_2)
R11
J3
Q1
Net-(Q1-G)

BOARD 3

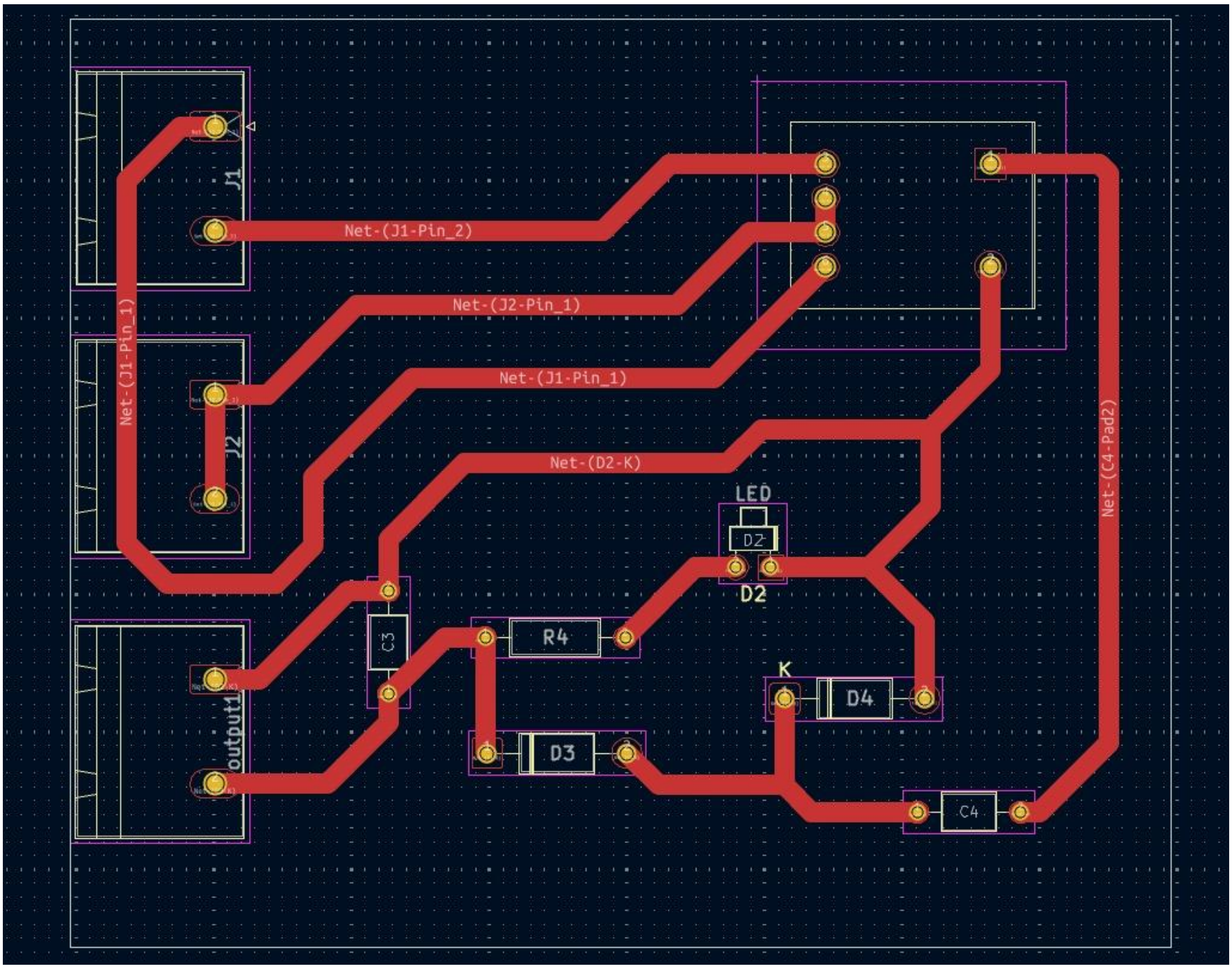
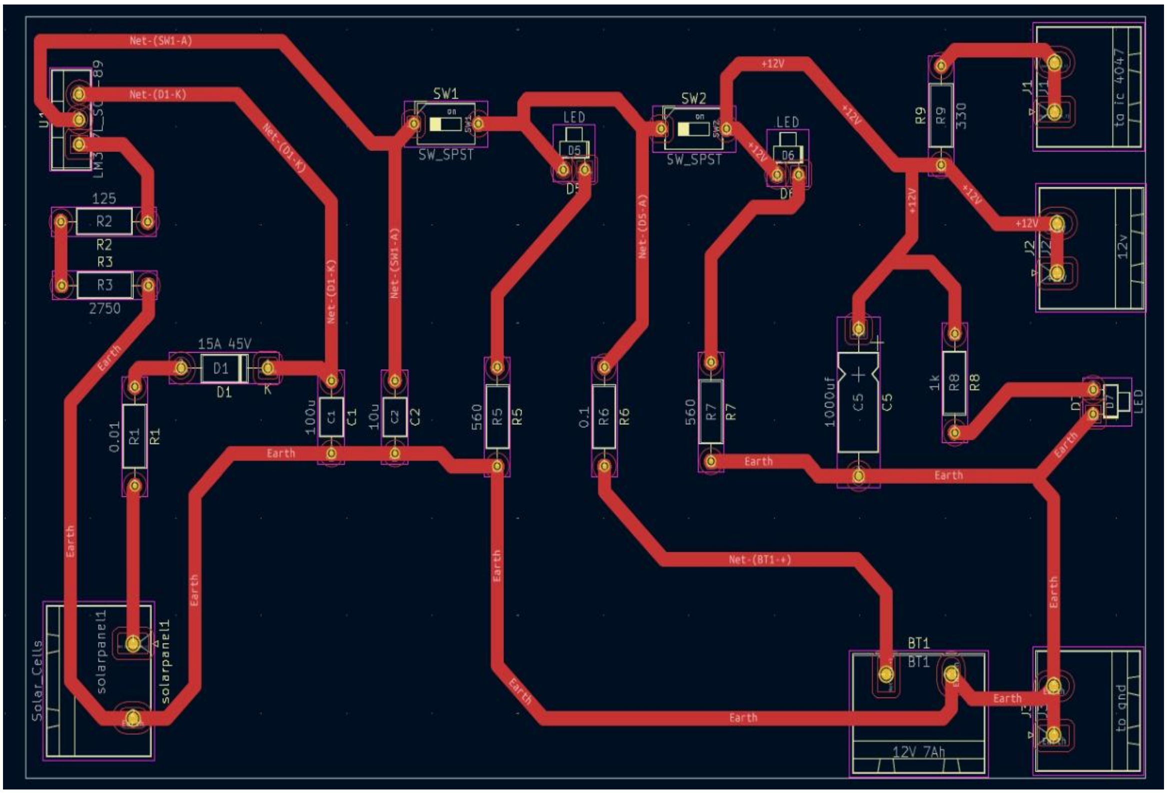


Figure 2: PCB layout

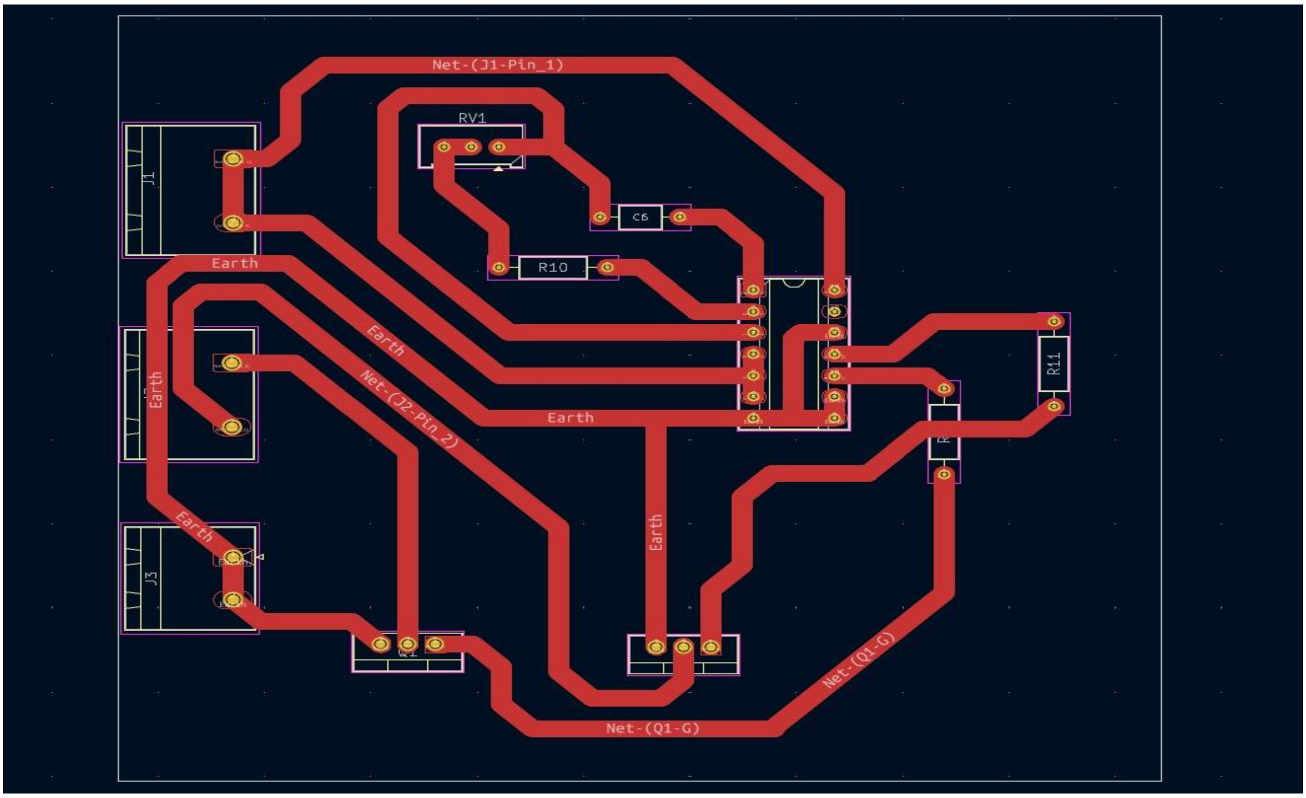
6. Front Copper

The front copper layer contains the main routing for power lines, signal traces, and pads for through-hole and surface mount components. Care was taken to ensure wide tracks for high current paths.

BOARD 1


Net-(SW1-A)
Net-(D1-K)
SW1
SW_SPST
SW2
SW_SPST
LED
+12V
R9
330
to ic 4047
J1
J2
12v
125
R2
R3
2750
15A 45V
D1
K
0.01
R1
100u
C1
10u
C2
560
R5
0.1
R6
R7
1000uf
C5
1k
R8
LED
Earth
Net-(BT1-+)
BT1
12V 7Ah
to gnd
solarpanel1
Solar_Cells

BOARD 2


Net-(J1-Pin_1)
RV1
C6
R10
J1
Earth
Net-(J2-Pin_2)
R11
J3
Net-(Q1-G)

BOARD 3

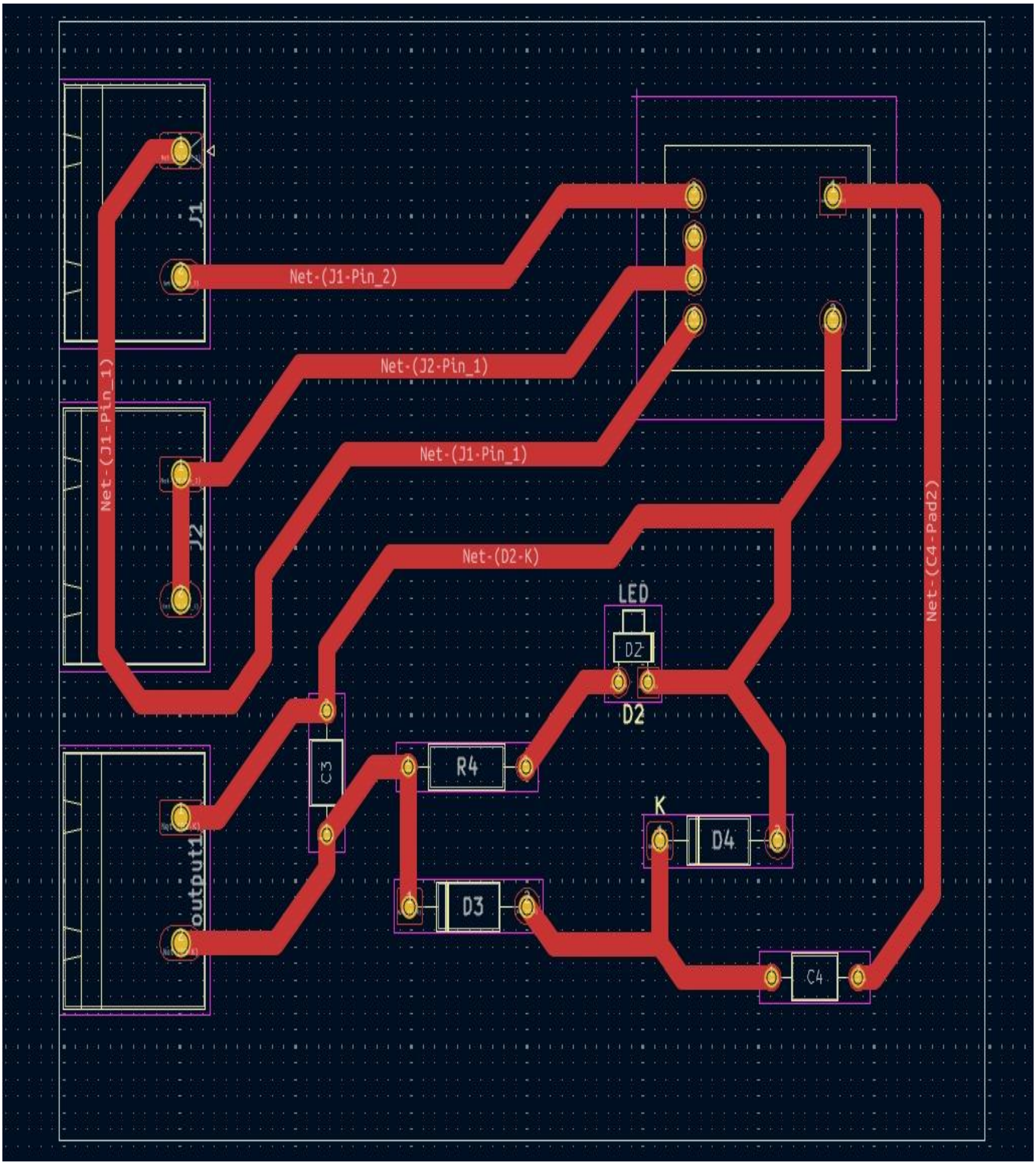
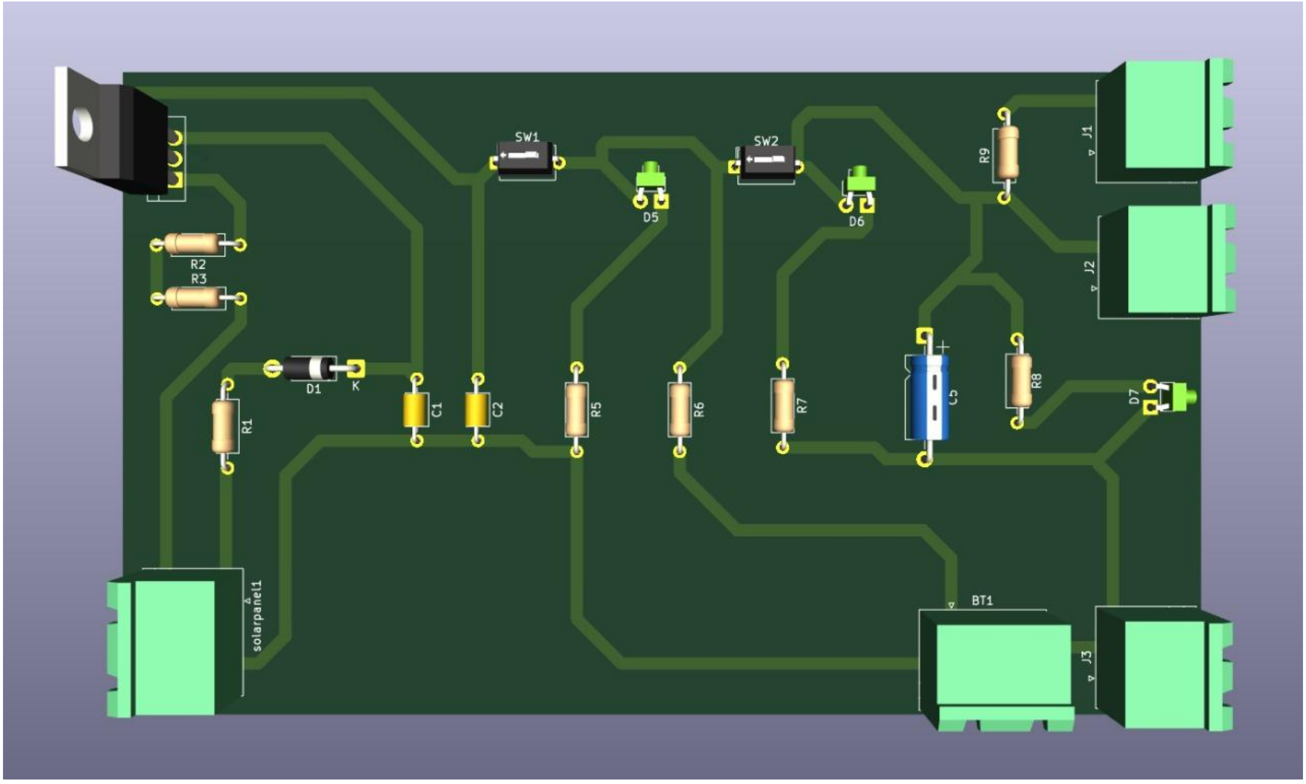
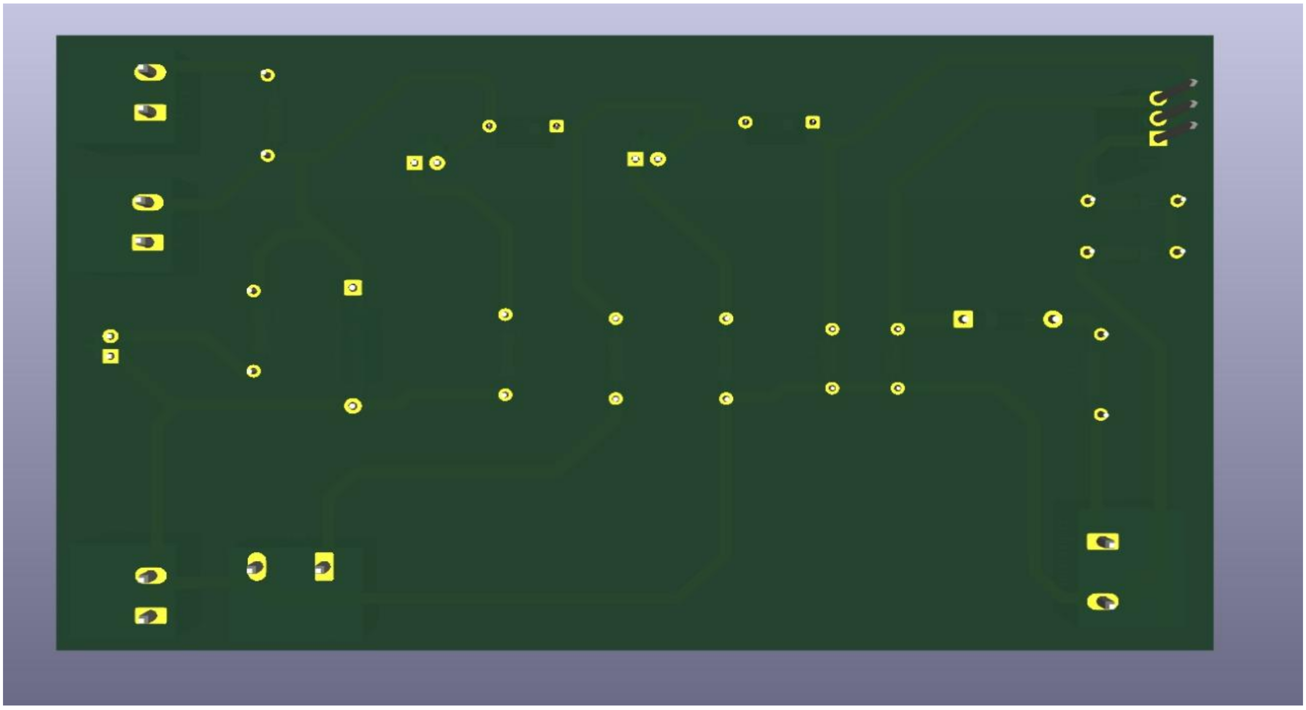


Figure: Front copper layer of the PCB.

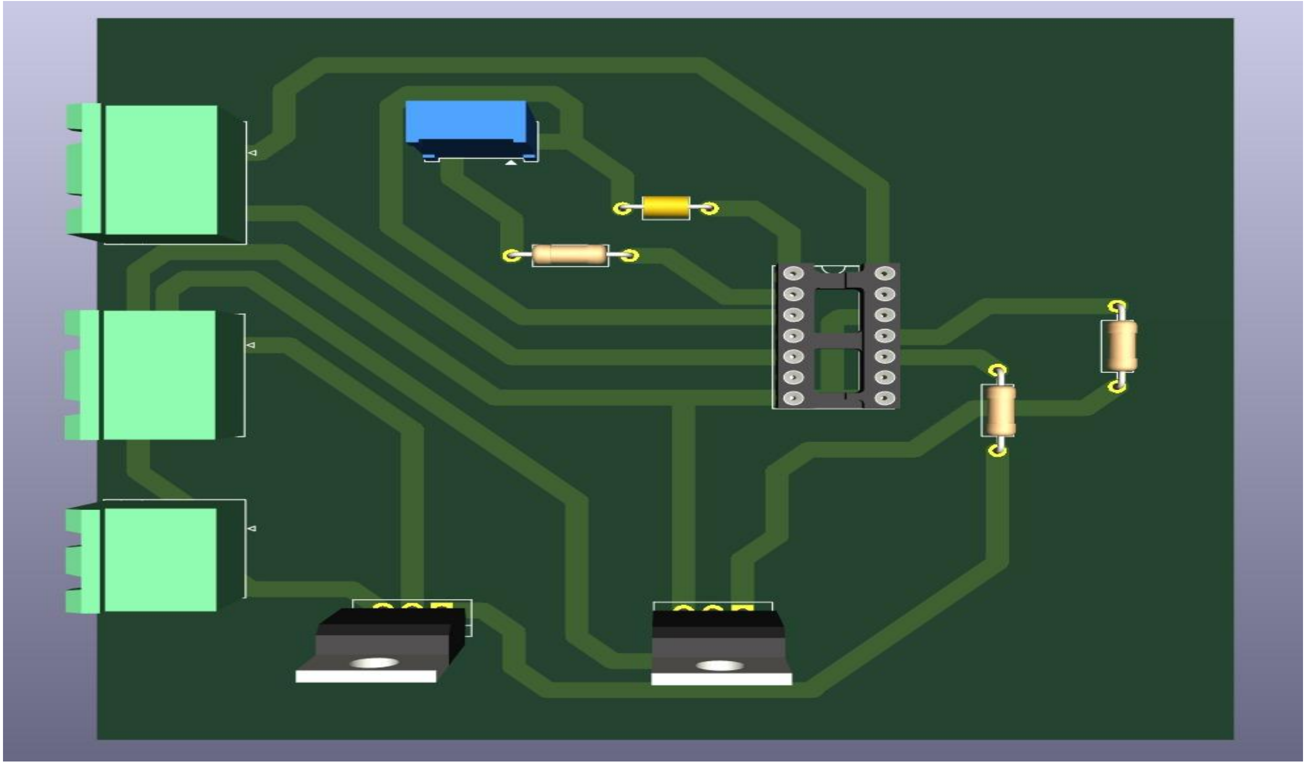
7. 3D View

Board 1

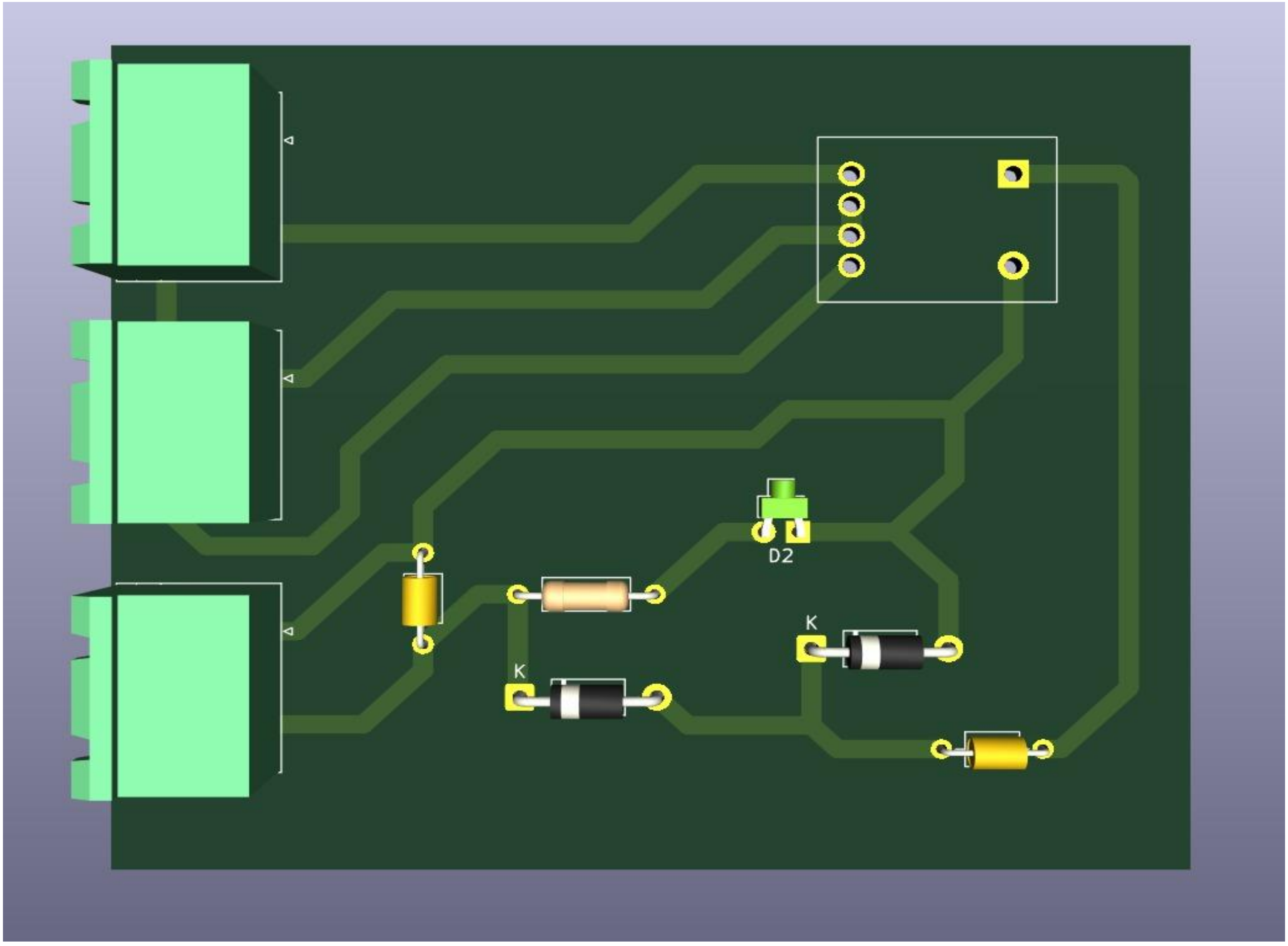


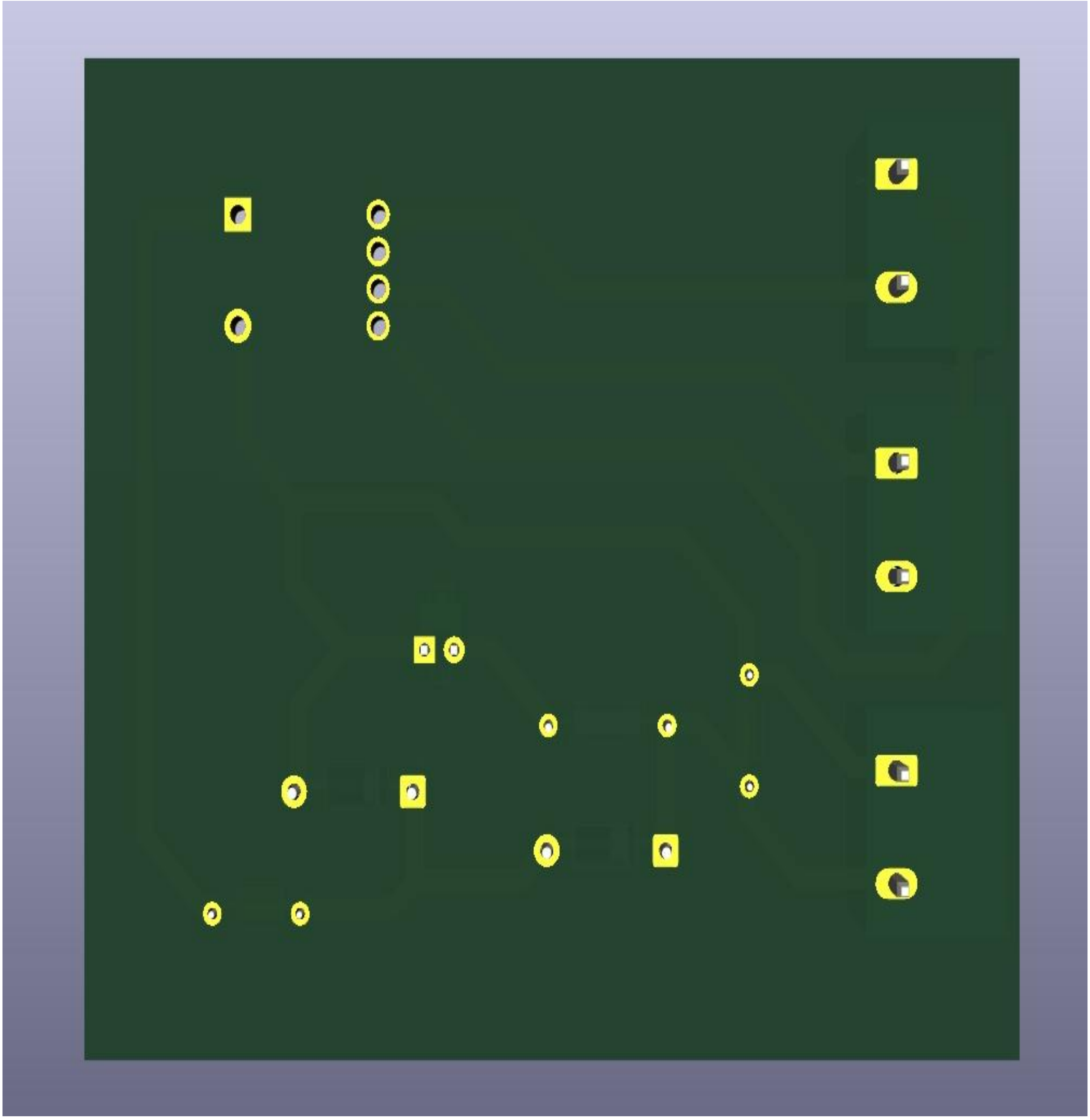


Board 2



Board 3

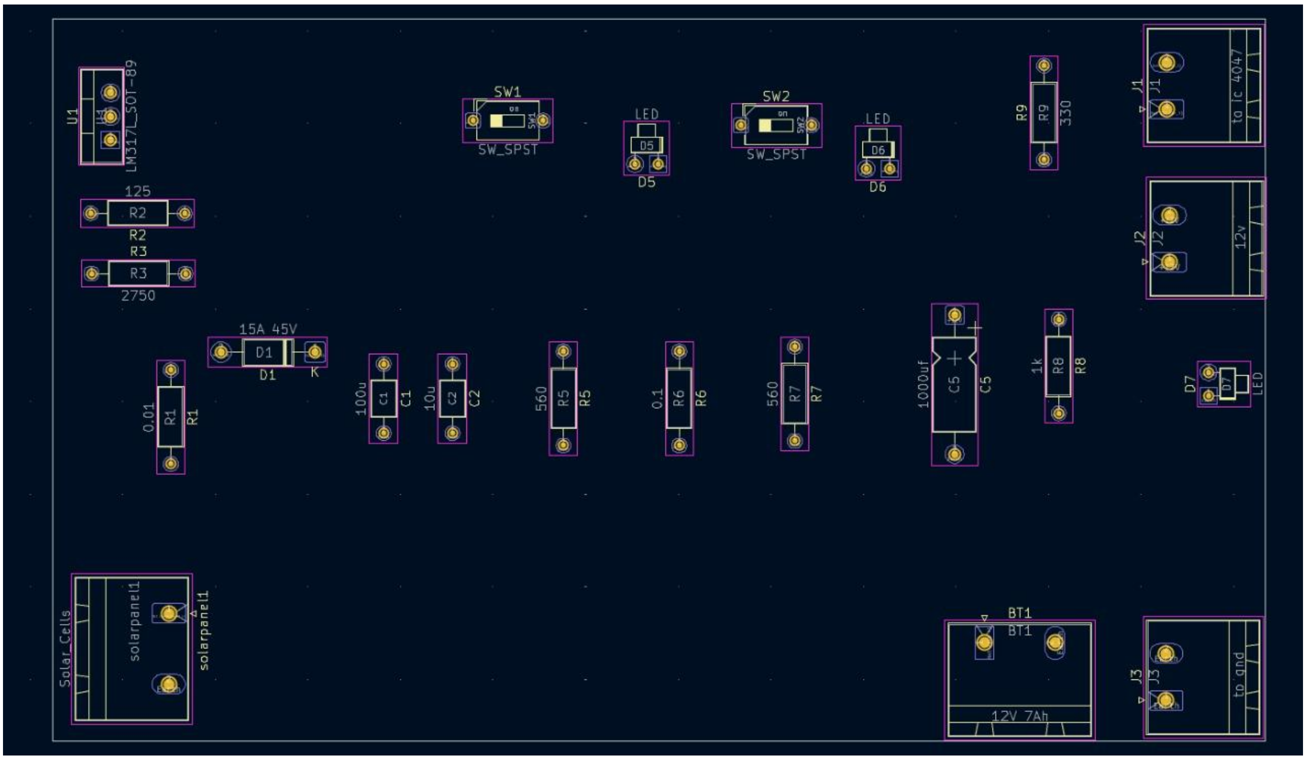




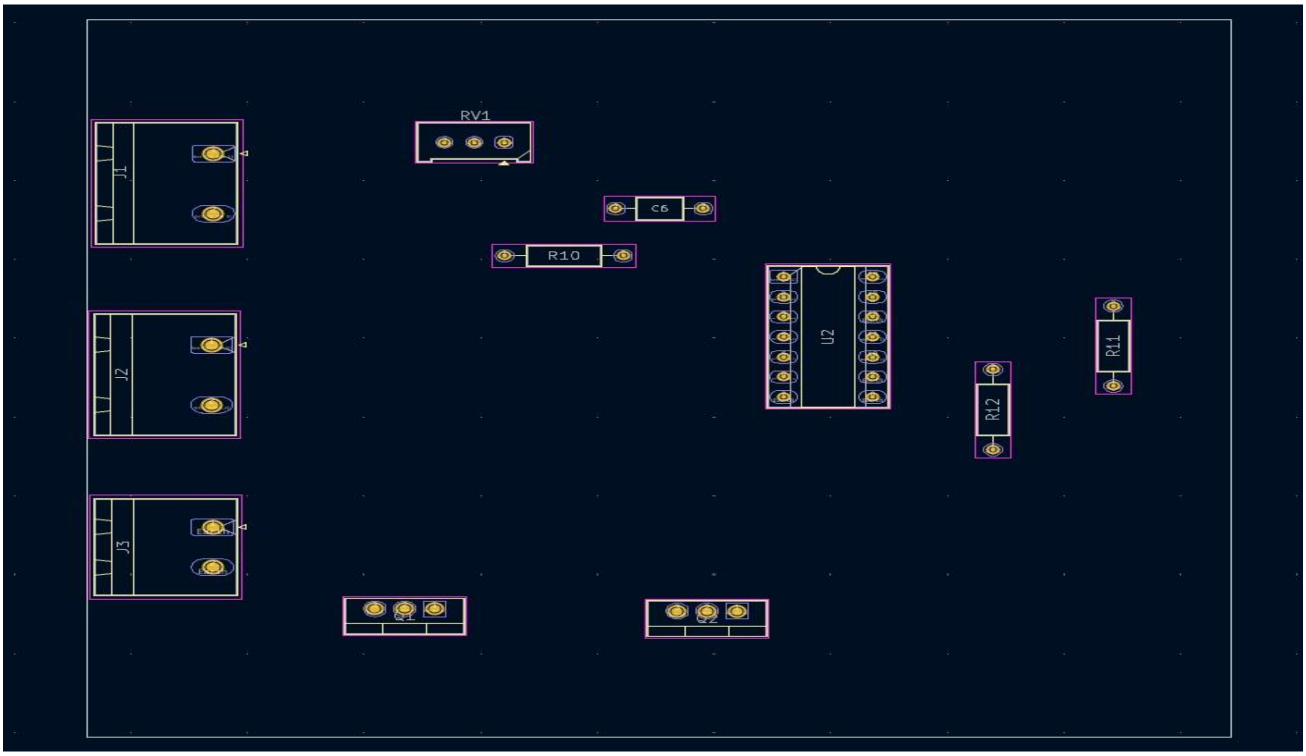
8. Legends

The silkscreen legends provide the component designators, polarity indicators, and assembly guides for accurate component placement and troubleshooting.

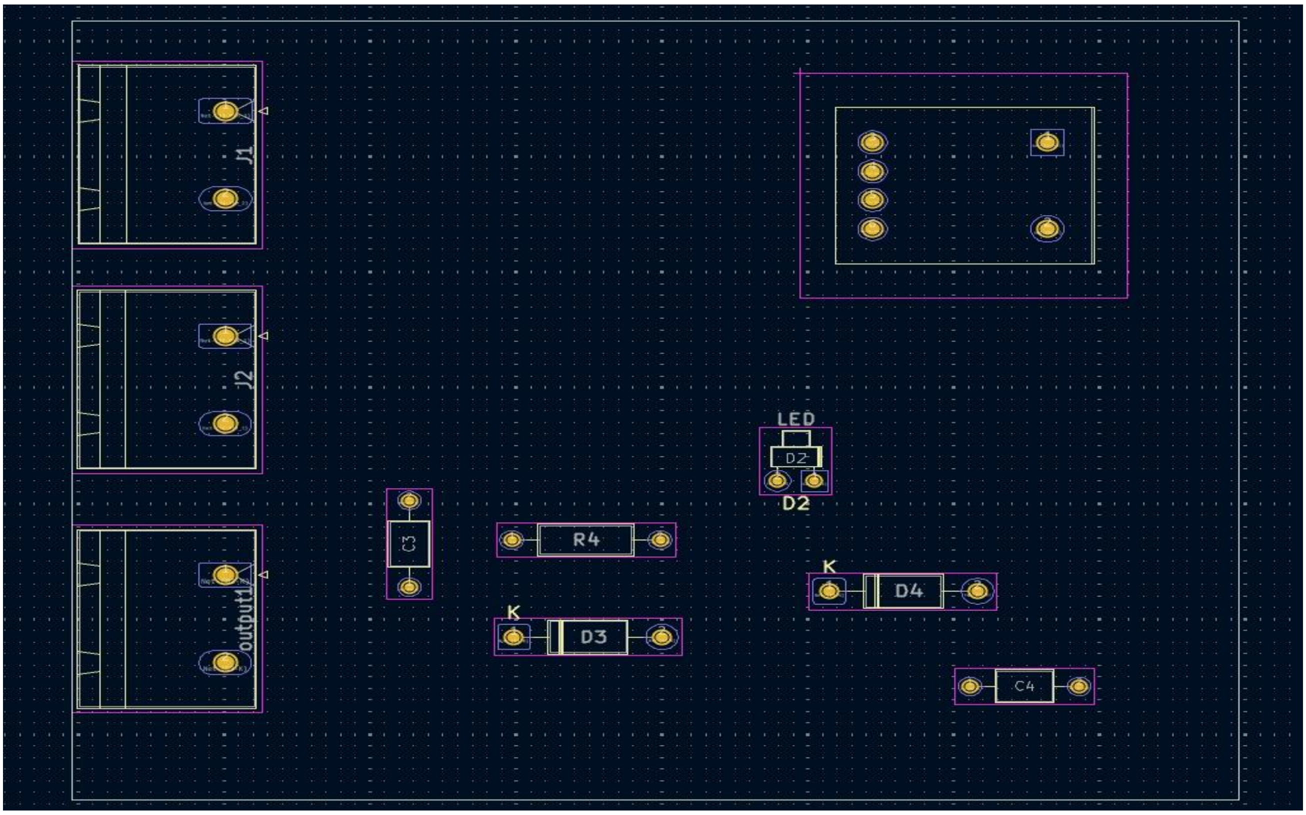
BOARD 1



BOARD 2



BOARD 3



Figures: Silkscreen Legend Layer of PCB

9. Overall Project Summary

The **Solar Powered Electric Fence** project demonstrates a sustainable and cost-effective solution for protecting agricultural fields and livestock using renewable energy. The system utilizes solar energy to power a fence that delivers controlled electric pulses, effectively deterring animals and intruders without causing harm.

The design integrates key electronic components such as the **solar panel, LM317L voltage regulator, CD4047 IC, MOSFETs, transformer, and protective diodes** to ensure reliable performance. The solar panel charges a **12V battery**, enabling continuous operation even during night time or cloudy conditions. The **CD4047 IC** generates pulses that switch the **MOSFETs**, which in turn drive the transformer to produce high-voltage pulses transmitted along the fence wire.

This project emphasizes **energy efficiency, safety, and practicality**. By operating completely off-grid, it offers a dependable fencing solution for remote and rural areas. The system's PCB layout, thermal management, and protective design ensure durability and ease of maintenance in outdoor environments.

Overall, the project successfully integrates **renewable energy, electronics, and safety engineering principles** to create an effective solar-powered electric fencing system. It serves as a valuable step toward **sustainable agricultural protection and energy-efficient rural technology**.